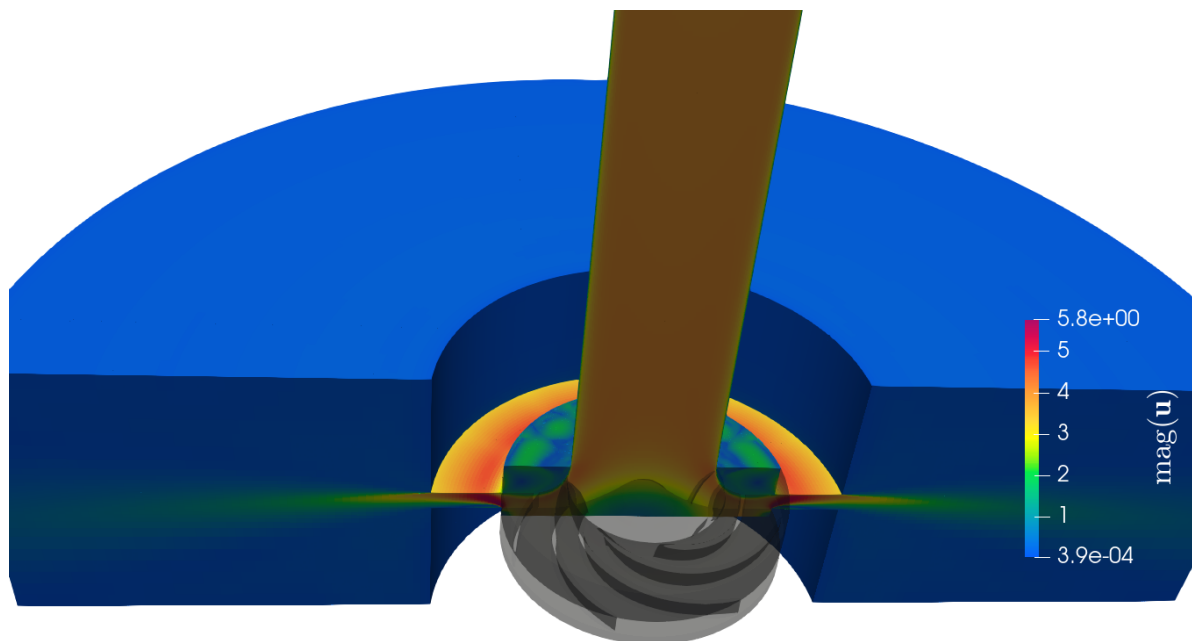


End-of-study thesis for engineering degree

Optimization of the performance of a centrifugal pump using Data Assimilation

Intern in Rotating Machines



Cross-section of the pump model

Educational tutor: Samy LABSIR

Laboratory tutors: Marcello MELDI, Antoine DAZIN, Miguel MARTINEZ VALERO

Clément THIBAULT - Aéro 5

From February 26th, 2025 to September 2nd, 2025

Version of July 10, 2025

Acknowledgments

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///// Changer la tournure pour éviter les répétitions de thanks... ///// Example : merci à toutes ces personnes : X pour..., y pour ...,...

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Summary sheet

The aim of this technical assessment is to summarize the gaps and their causes between the course's missions and what has been achieved. It also enables us to suggest what could be done by the next person working on the same subject.

Summary sheet		Clément THIBAULT - Aéro 5 TIE	
Internship topic		Objectives	
Optimization of the performance of a centrifugal pump using Data Assimilation		Optimize the forcing parameters of an IBM ¹ in a centrifugal pump using DA with CONES, to find the correct performance of the pump.	
Main customer		Used tools	
- LMFL - ENSAM		OpenFoam 9, CONES, VSCode, LaTeX, Git, Python,C++, Bash, TGCC, Paraview	
Conducted studies			
- Coding the DA test case in OpenFoam and CONES - Convergence of the CFD solution (physical and numerical stability/precision) - Convergence of the EnFK, adding tools as inflation			
Results		Explanation of possible differences	
- Drop up to 50% of the mathematical NRMSE with DA (almost 100% expected) - Implementation of inflation, in CONES - Implementation of radial expansion, "azimutal interpolation" in OpenFoam 9 - Discovering of a recirculation bubble in the diffuser		- After the diffuser, the impeller's wake direction is reversed because of the MRF method - The rotating impeller shape combined with the IBM allows non-physical flow that induce a recirculation bubble. - Results of the EnKF are not as good as expected because of the IBM method combined to MRF method, which is not adapted to the centrifugal pump geometry.	
Encountered difficulties		Work to be continued	
- Understanding the tools OpenFoam and CONES - Code everything directly on the TGCC terminal - Initialize physical field of complex instationary simulation		- Use the same type of DA for the analysis of axial compressor internal flows - Synchronize PIV and simulation in the DA - Implement interpolation IBM method	

Note: Acronyms are define later.

¹All acronyms are defined from the next page or available in the glossary.

General Introduction

Since I was young, I've always been fascinated by fluid mechanics. I remember watching planes flying and being amazed by the way they could fly. I was also interested in the way water flows in rivers and sailing boats. In high school, I began working on plane-related projects and explored them more theoretically through Computation Fluid Dynamics (CFD).

This fascination led me to study physics and mathematics at university, where I discovered applied mathematics including CFD, Artificial Intelligence (AI) and Kalman Filter. During my studies, I've become passionate about applied mathematics and fluid mechanics. That's why I did my last internship at the Centre Européen de Recherche et de Formation Avancée en Calcul Scientifique (CERFACS). It convinced me to do a thesis in CFD after Institut Polytechnique des Sciences Avancées (IPSA), my current engineering school.

The subject proposed by the Laboratoire de Mécanique des Fluides de Lille (LMFL) really suited me: CFD using an open-source library, coupled with AI and experimental data. So it was with enthusiasm that I started this internship on February 26, 2025, in this laboratory for a duration of six months.

Briefly, the LMFL is a public laboratory, consisting of 38 researchers and around 25 PhD students. The laboratory is present on three sites located in Lille (École Nationale Supérieure d'Arts et Métiers (ENSAM) and Office National d'Études et de Recherche Aéronautiques (ONERA)) and on the Villeneuve d'Ascq campus (Centrale Lille, Centre National de la Recherche Scientifique (CNRS), University of Lille).

My main tasks during this internship were:

- Ensuring the convergence of the CFD solution;
- Implementing the Data Assimilation (DA) problem in OpenFOAM and Coupling OpenFoam with Numerical EnvironmentS (CONES);
- Optimizing the forcing parameters of the Immersed Boundary Methods (IBM) using DA with CONES to correct a non-physical behavior.

To present the details of my internship work, I will first introduce the LMFL research laboratory. In the second part, I will describe the work I carried out, structured into the following five sections:

1. Introduction to DA problems;
2. Numerical ingredients of the CFD and DA problems;
3. Preliminary results using IBM;
4. Application of DA to the IBM;
5. Results and perspectives.

Chapter 1

The LMFL laboratory and Arts et Métiers Lille campus

1.1 Scientific environment and host institution

The LMFL is a public research laboratory jointly supervised by Centrale Lille, Université de Lille, Arts et Métiers ParisTech, ONERA, and the CNRS.



Figure 1.1: LMFL facilities; Top: Villeneuve d'Ascq campus, bringing together Centrale Lille, the CNRS, and Université de Lille. Bottom-left: ONERA. Bottom-right: Arts et Métiers campus (ENSAM) (source : lmfl.cnrs.fr)

Over the past six months, I have been hosted by ENSAM, specifically at the Lille campus. Arts et Métiers is a prestigious engineering school focused on mechanical, industrial, and energy engineering. The Lille campus to students a dynamic and innovative environment in a region with a strong industrial heritage.

The LMFL was created on January 1, 2018, from the merger of two research units: the “Écoulements Tournants et Turbulents” team from the Lille Mechanics Laboratory, and the “Expérimentation et Limite de Vol” unit. This diversity of expertise enables the research teams to engage in numerous collaborative projects with regional and national partners.

Research conducted within the LMFL is organized into three main topics:

- **Turbulence** : Studying and modeling turbulent flows using numerical and experimental methods. The focus is on inhomogeneity and non-stationarity in wall-bounded flows, wakes, jets, and various other flow configurations. This research area includes 5 permanent researchers and around 10 PhD and postdoctoral students.
- **Rotating flows** : Analyzing and modeling internal and external flows related to rotating machines, such as turbo-machinery and rotors. Experimental and numerical studies of rotating machines (turbomachines, rotors, propellers) operating under degraded conditions, as well as flow control, cavitation, and multiphase flows, are also key topics. This subject gathers 6 researchers, 7 research engineers and about 13 PhD and post-doctoral students.
- **Flight dynamics in unsteady and non-uniform environments** : Development of analytical and experimental methods to determine the evolving behaviour of low-speed aircraft in perturbed flight domains. Post-stall dynamics, environmental interactions, and experimental design are among the main research interests. This topic involves about 5 research engineers and 5 PhD and post-doctoral students.

1.2 Social and environmental responsibility

ENSAM's Corporate Social Responsibility (CSR) (or Responsabilité Sociétale des Entreprises (RSE)) action plan is structured around five strategic axes¹²:

1. **Strategy, governance and territorial anchoring**: presents ENSAM's actions in terms of sustainable development, in particular the Evolutive Learning Factories 4.0 project aimed at modernizing training and research platforms by fully integrating the CSR dimension.
2. **Education and training**: Training students in eco-responsibility
3. **Research and innovation**: Ensure that all research themes have direct or indirect impacts on the achievement of international and national sustainable development goals. Ensure free access to research results and data.
4. **Environmental management**: ENSAM aims to reduce its environmental impact by improving energy efficiency, cutting emissions (including Scope 3), and integrating carbon accounting. It also promotes biodiversity and sustainable mobility through staff engagement, responsible purchasing, and green-space management.
5. **Social policy**: Reconciling economic, societal, and environmental dimensions by training staff, teachers, research professors, and stakeholders in sustainable development issues, developing a quality of life policy within the institution, and ensuring equal opportunities in training, research, and human resources. An indicator is the QVT, that is detailed in the Rapport Social Unique (RSU)³ 2024.

The "Comité Social et Économique (CSE)" is the staff representative body within ENSAM. Although specific information on the CSE is limited to the 2022 RSU mentioning the monitoring of the actions of the "Comité d'Hygiène, de Sécurité et des Conditions de Travail National (CHSCTN)" which is the old name of the CSE.

Arts et Métiers also publishes its activity reports, its social balance sheet / single social report and public data then transmitted to the Commission des Titres d'Ingénieur (CTI) available online

¹<https://artsetmetiers.fr/fr/ecole-engagee>

²https://dp-www.s3.ensam.eu/public/2024-11/24_032_plaquette-bilan-RSE_V9_0.pdf

³https://dp-www.s3.ensam.eu/public/2025-07/RSU%202024_VCA.pdf

⁴. More specifically, students from Arts et Métiers created a webpage to learn about general CSR and sustainable development: ⁵ (addressing themes such as ecology, inclusion, gender equality and the fight against discrimination,...).

About the LMFL we can also find an interesting documents from Haut Conseil de l'Évaluation de la Recherche et de l'Enseignement Supérieur (HCERES). This organization is in charge of evaluating all research structures. Every five years, it publishes a report on the LMFL. The last one⁶ was published in February of this year. It presents the laboratory's strengths and weaknesses, as well as its research activities, collaborations, workforce, and future perspectives. It highlights the laboratory's commitment to excellence in research, its multidisciplinary approach, and its contributions to the field of fluid mechanics and related areas. More precisely, the report emphasizes the laboratory's strengths in experimental and numerical methods, its expertise in turbulence and rotating flows, and its contributions to the understanding of complex fluid phenomena.

The report highlights that:

- The facilities are unique, with wind tunnels and turbomachinery test rigs that allow advanced experimental investigations of external and internal flows.
- It is one of the most active European laboratories in the field of next-generation aircraft design, particularly on electric propulsion systems, supported by high-level experimental setups such as the Eiffel wind tunnel (L1) and a vertical wind tunnel with a rotating balance. The LMFL also leads European research on airship aerodynamics and urban drone flight.
- Its experimental platforms are completed by advanced numerical simulations, with equipment partly acquired through regional Contrat de Plan État-Région (CPER) programs. The lab conducts original research on turbulence dissipation mechanisms in non-homogeneous, unsteady flows and on turbulent/non-turbulent interfaces.
- Efforts have been made to create synergy between the lab's three geographically separated sites through transversal research themes in optical metrology, flow control, and data analysis. Weekly LMFL webinars support internal collaboration.
- Despite some staff departures, LMFL has maintained its attractiveness by recruiting top researchers, strengthening historical themes like turbulence and instabilities, and expanding into areas like two-phase aerodynamics and data-driven modeling.
- LMFL's scientific output is of high quality, with publications in top journals (e.g., Journal of Fluid Mechanics (JFM)) and participation in major conferences. While flight dynamics publications are fewer, this area is expected to grow through collaborations, particularly with the turbulence team.
- The lab also benefits from significant external funding via competitive calls (ERC, ANR, Clean Sky) and strong industrial partnerships (e.g., CNES, Ariane Group). The unit's leadership has been commended for its organizational work and encouraged to keep fostering synergy across research themes.

⁴<https://artsetmetiers.fr/fr/observatoire-des-donnees>

⁵<https://artsetmetiers.fr/fr/actualites/developpement-durable-et-rse-faites-le-plein-de-connaissances-sur-savoir>

⁶https://www.hceres.fr/sites/default/files/media/publications/rapports_evaluations/pdf/

My personal experience of the work environment has been very positive. The working atmosphere within the laboratory is pleasant and conducive to research. The researchers are passionate about their work and share their knowledge with enthusiasm. The team is close-knit, which encourages the exchange of ideas and collaboration between members from all over the world. The working atmosphere suits me: researchers generally work hard, but there is no pressure regarding working hours. The building is a beautiful old, typical Lille building, but well renovated and adapted to the research work being conducted (both numerical and experimental). The laboratory offers spacious areas, interior courtyards, large lecture halls, and a large workshop for experiments.

Chapter 2

Optimizing the performance of a centrifugal pump by data assimilation

The internship began in February 2025. I was provided with a fix computer, credentials to connect to the Irene and Cassiopée supercomputers, and the link to the CONES's gitlab¹. Marcello MELDI, Antoine DAZIN and Miguel MARTINEZ VALERO which are my internship supervisors, showed me around the lab. Then, Tom MOUSSIE and Miguel, my co-officies, explained to me the basics of OpenFoam and CONES, the DA code. I was able to run a few tests on OpenFoam on the supercomputers, which allowed me to familiarize myself with the tools and the code.

J'aimerais ajouter une introduction "pourquoi ce sujet de stage existe ?" (où l'ajouter ?)

2.1 Data Assimilation

DA[2] is a technique used to improve the accuracy of numerical models by combining them with observational data. It is widely used in various fields, including meteorology, oceanography, and engineering. The goal of data assimilation is to obtain a more accurate representation of the state of a system by integrating information from different physical measurements or high fidelity sources.

2.1.1 State of the art

It does exist many methods for data assimilation, distributed in three main categories: **variational methods**, **statistical methods** and **hybrid** (see Figure 2.1). Variational methods (also called classical methods), such as the 3D-Var and 4D-Var, use optimization techniques to find the best estimate of the state of a system by minimizing the difference between model predictions and observations. Statistical methods (also called sequential methods), such as the Kalman filter and its variants (e.g., EnKF), update the model state iteratively as new observations become available.

¹<https://gitlab.ensam.eu/pe431/cones-dev>

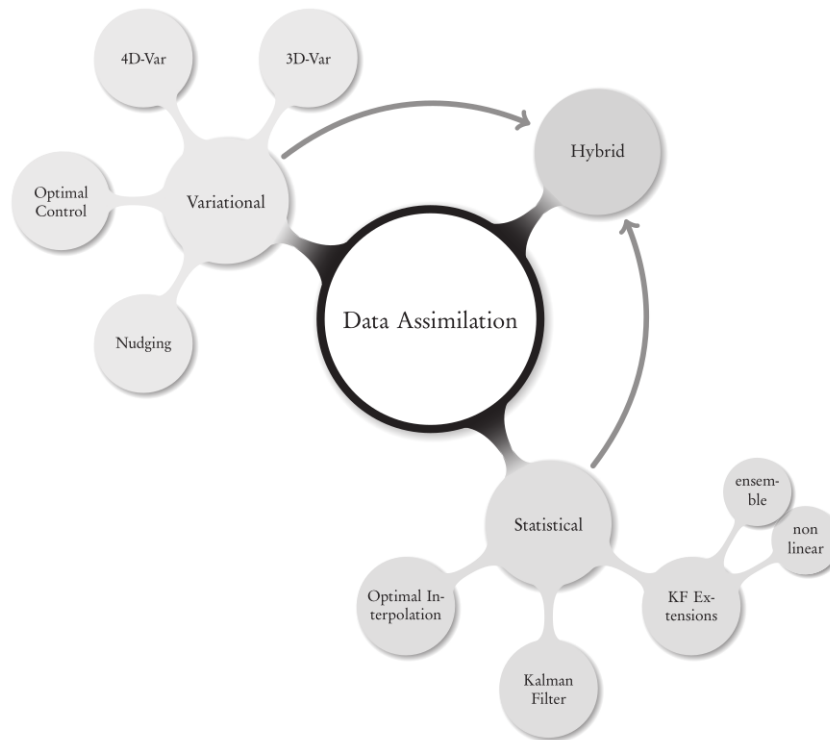


Figure 2.1: The big picture for DA methods and algorithms[2]

EnKF Our approach to data assimilation is based on the Ensemble Kalman Filter (EnKF), which is a Monte Carlo method that uses a set of model states (the ensemble) to estimate the state of the system. The EnKF is particularly well-suited for nonlinear systems and has been successfully applied in various fields, including computational fluid dynamics (CFD).

EnKF for DA in CFD has been used at LMFL since a few years, with the work of Lucas Villanueva[10], Miguel M.Valero[10], Tom Moussie[5], Sarp Er, Paolo Errante[5] and Marcello Meldi[10][5] and his colleagues. The EnKF method is a powerful tool for assimilating data into computational fluid dynamics (CFD) simulations, allowing for the correction of model predictions based on observed data. An illustration of the EnKF data assimilation process is shown in Figure 2.2. The maths behind the EnKF method is detail in the subsection 2.2.3

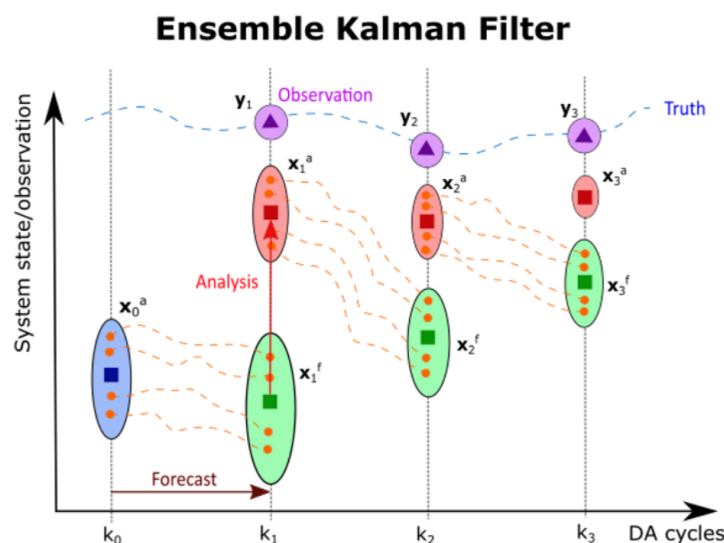


Figure 2.2: Illustration of the EnKF process (100% confidence in observations)

CFD Today, we can theoretically run a Direct Numerical Simulation (DNS) which resolves exactly the Navier-Stokes equations, but it is still too expensive to be used in practice. Therefore, we generally use Large Eddy Simulation (LES) or Reynolds-Averaged Navier-Stokes equations (RANS) models to simulate the flow. The EnKF method can be applied to these models to improve some of them parameters or physical state, to improve the physical accuracy.

A good example based on the simple and academic *cavity test case* illustrate it well. The cavity test case is a simple flow in a square cavity, where the flow is driven by a moving lid. The flow is characterized by a recirculation zone and a vortex in the center of the cavity. The EnKF method can be used to assimilate data from the flow, such as velocity or pressure measurements, to improve the accuracy of the simulation. The EnKF method can also be used to estimate the parameters of the model, such as the viscosity or the turbulence model coefficients, to improve the physical accuracy of the simulation.

In our case, we are going to optimise the velocity wall condition. For that, we use the result on 5 locations of a precise converged simulation with a top boundary condition of 5 m.s^{-1} as a reference. We will then assimilate data from the flow, at the 5 observation points (see Figure 2.3a). We start from a simulation with a wall condition of 1 m.s^{-1} , which is not the correct value. The EnKF method will then be used to correct the wall condition by assimilating the data from the flow at the 5 observation points. The EnKF method will iteratively update the wall condition until it converges to the correct value, which is 5 m.s^{-1} in this case. In only 3 EnKF iterations, the wall condition converges to the correct value (with a relative error of 0.05 m.s^{-1}).

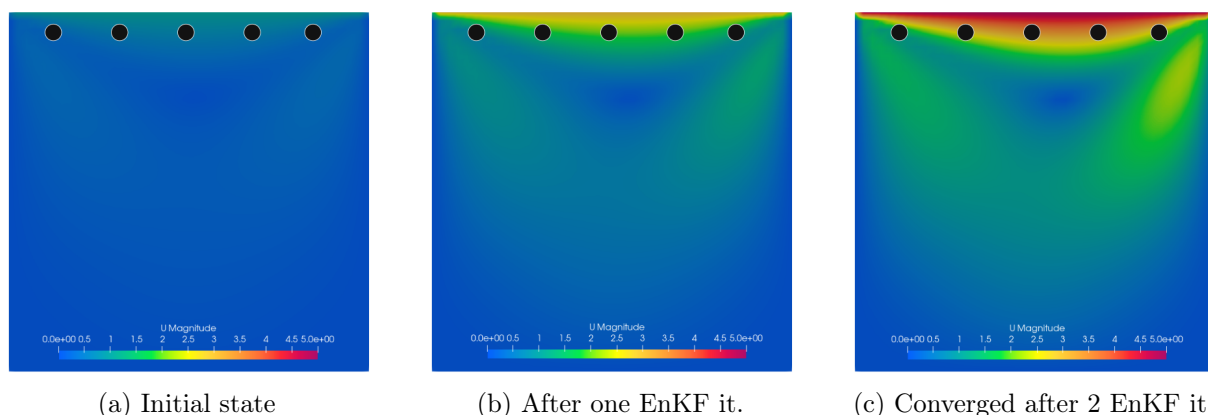


Figure 2.3: Evolution of the state through EnKF iterations (●: observation point)

IBM The IBM theory has been developed in 1972 by Charles S. Peskin[6]. The IBM is a numerical technique used to simulate fluid-structure interactions, particularly in cases where the structure is immersed in a fluid flow. It permit to keep numerical precision on complex geometries and moving boundaries without the need for body-fitted meshes, which can be computationally expensive and difficult to implement.

They are different ways to discretize the space when we do CFD (see Figure 2.4²). The most common are structured meshes, unstructured meshes and body-fitted meshes. The body-fitted meshes are the most accurate but also the most expensive in terms of computational resources. The IBM allows to use unstructured meshes, which are less accurate but more efficient in terms of computational resources.

²https://www.solidworks.com/sw/docs/flow_basis_of_cad_embedded_cfd_whitepaper.pdf

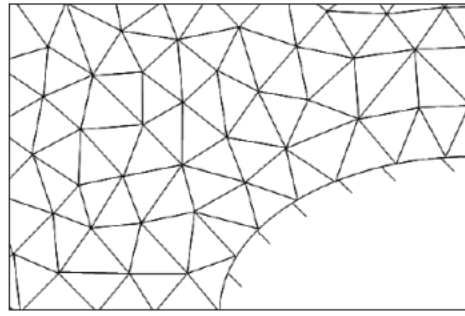


Figure 1: Unstructured body fitted mesh

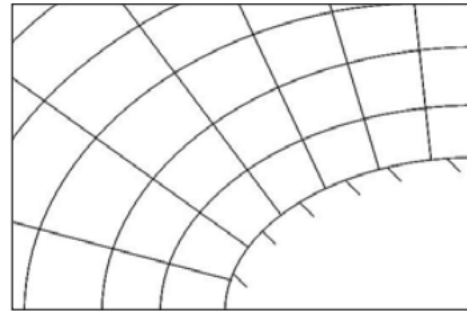


Figure 2: Structured body-fitted mesh

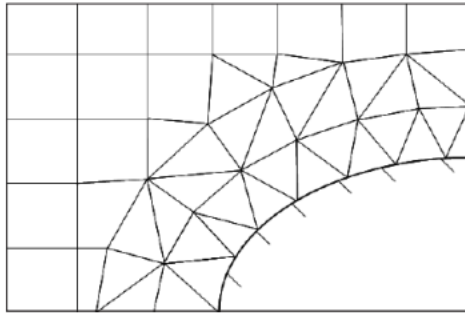


Figure 3: Combination of structured Cartesian mesh and non-structured body-fitted mesh near the wall

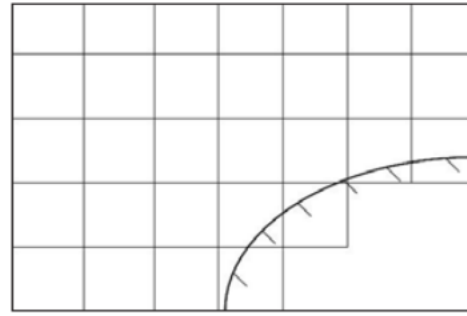


Figure 4: Structured Cartesian immersed-body mesh

Figure 2.4: Different ways to mesh

In our case, we impose a forcing term on the Navier-Stokes equations to simulate the effect of the immersed boundary on the fluid flow. This term is added only on the penalization part $\Omega_b U \Sigma_b$ (see Figure 2.5). The maths behind the EnKF method is detailed in the subsection 2.2.1. The problem of this method is that it leads to a few constants that need to be tuned to obtain the best results. For that, we use the EnKF method to optimize these constants. The EnKF method will be used to assimilate data from the flow at the observation points and to update the forcing term until it converges to the optimal value.

These high fidelity data are obtained from an experiment explained in the next subsection 2.1.2

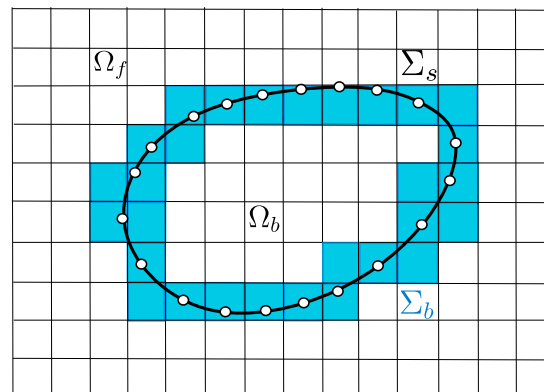


Figure 2.5: IBM regions

This year, my supervisors M. VELRO and M. Meldi published an article[8] about the optimisation of the IBM forcing and turbulence terms, that gives very good results. More recently, they obtain very good results on the optimisation of these parameters on an oscillating cylinder. //
Il faut ça à vérifier en relisant la thèse de Miguel. //

This leads us to the idea of using the EnKF method to optimize the parameters of the IBM forcing term on a centrifugal pump, which is a more complex case than the previous ones. The

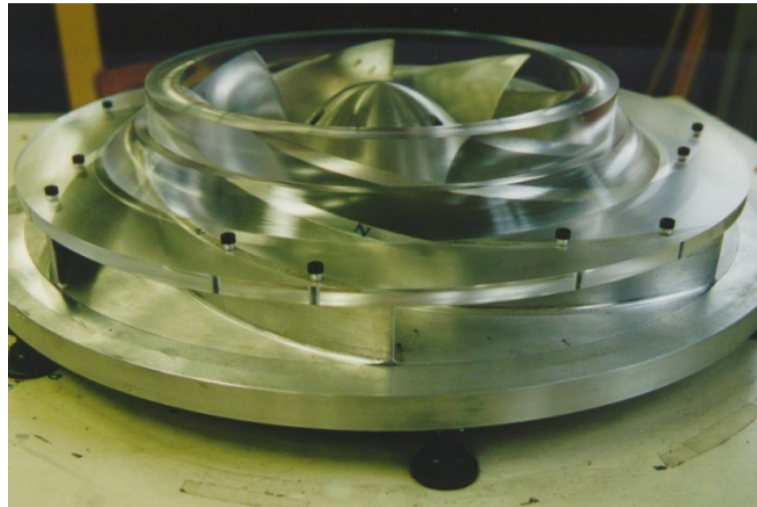
goal is to improve the performance of the pump by optimizing the parameters of the IBM forcing term, which will lead to a better simulation of the fluid flow and a better understanding of the pump's behaviour.

2.1.2 Presentation of the data from the centrifugal pump test case

The pump Centrifugal pumps are used since 1475[7] and are still widely used today in various applications, such as water supply, irrigation, turbomachines, and industrial processes. They are known for their low cost, efficiency and reliability, making them a popular choice for many fluid transport applications. The studied centrifugal pump is a Société Hydrotechnique de France (SHF) impeller, which is a type of centrifugal pump designed for high flow rates and low head applications.



(a) Old centrifugal pump used to move hay (personnal picture)

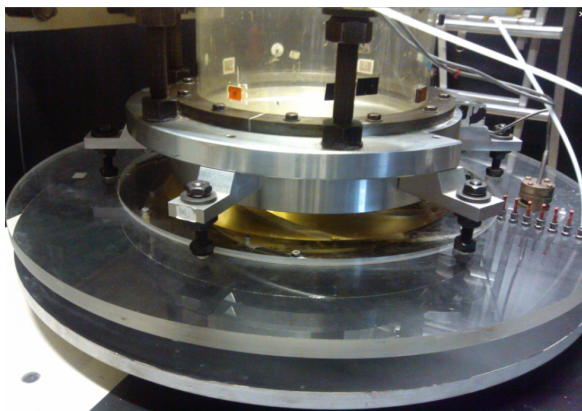


(b) SHF impeller used in the RESEDA benchmark

Figure 2.6: Centrifugal pumps

Different high fidelity data has been recorded on the diffuser of the pump at the LMFL:

- High-Speed Particle Image Velocimetry (PIV) data obtained by Antoine Dazin [3] in 2011;
- Pressure data obtained by Meng Fan [4] in 2022.



(a) Vaneless diffuser of the centrifugal pump



(b) Pressure taps on the diffuser

Figure 2.7: Diffuser and pressure taps of the centrifugal pump

The operating conditions of the machine where :

Impeller characteristics		
R_1	Tip inlet radius	141.1 mm
R_2	Outlet radius	257.5 mm
Z	Number of blades	7
Q_d	Design flow rate	$0.236 \text{ m}^3/\text{s}$
K	Mean blade thickness	9 mm
Ω	Rotating velocity	1 200 rpm
β_2	Outside blade angle (from peripheral velocity $U = R_2\Omega$)	22.5°
Vaneless diffuser characteristics		
b_3	Diffuser width	38.5 mm
R_3	Inlet radius (without leakage, i.e., $R_3 = R_2$)	257.5 mm
R_4	Diffuser outlet radius	385.5 mm
t	Thickness of the upper and lower sections of the diffuser	20 mm

Table 2.1: Summary of the main geometrical and operating parameters of the impeller and diffuser at design conditions, employed in the numerical modelling of the centrifugal pump.

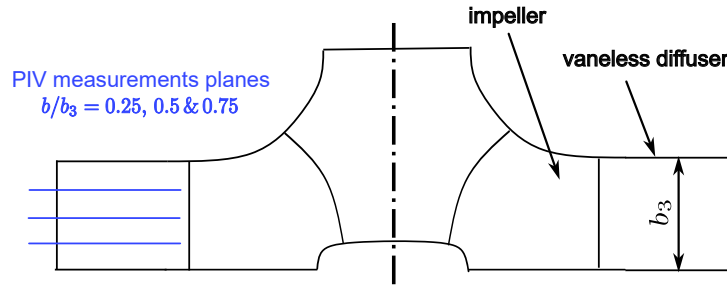


Figure 2.8: Position of the PIV plans on the diffuser

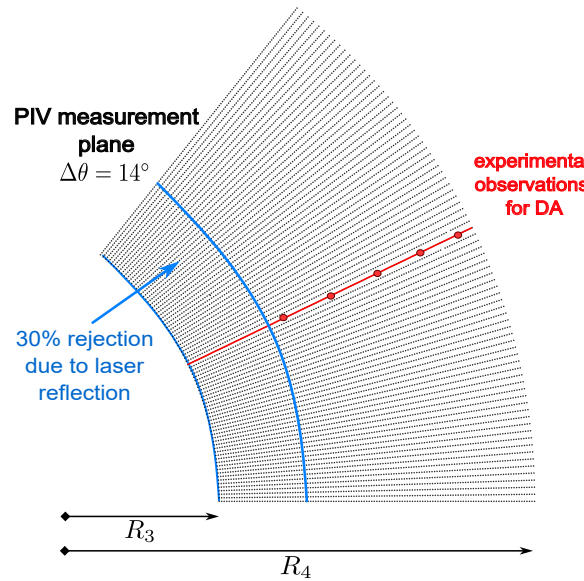


Figure 2.9: Position of the PIV points on a polar slice of the diffuser

PIV data The data are stored in a directory structure that contains six directories. The six directories are corresponding to two tests realised at 25%, 50% et 75% of the diffuser height. Each test was repeated twice for the same operating condition and the same height (R1, R2)

In each directory, there are 1581 files corresponding to instantaneous ASCII data file. The ten columns are corresponding to :

$$x(m), y(m), V_x(m.s^{-1}), V_y(m.s^{-1}), r(m), \theta(rad), Vr(m.s^{-1}), V_\theta(m.s^{-1}), V_z(m.s^{-1}), Unknown$$

And the 10,125 lines corresponds to the 81 locations over x and 125 locations over y of the PIV. Processing of these data are detail in the section 2.4.1.

Pressure data In addition to the nine pressure tapes on the diffuser, M. Fan has also capture the inlet and outlet pressure that allow to compute the global performance of the pump (see //partie L.N.M à référencer//)

//The theoretical average radial speed at the input of the diffuseur is equal to

$$\frac{Q_{dif_{in}}}{S_{dif_{in}}} = \frac{Q_{pump_{in}}}{2\pi R_3 h_{dif}} = 3.65 m.s^{-1}$$

With $Q_{pump_{in}} = Q_{dif_{in}}$ because the mass flow rate is constant and the flow is incompressible.//
-> No need ??

2.2 Numerical ingredients

2.2.1 CFD governing equations

La CFD est basée sur les équations fondamentales qui décrivent la mécanique des fluides newtoniens: les équations aux dérivées partielles non linéaires de de Navier-Stokes.

Elle regroupe les 3 principes physiques suivants, donnés ici dans une formulation eulérienne (explquer pk) :

- Conservation de la masse, avec l'équation de continuité :

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

incompressible :

$$\nabla \cdot \mathbf{u} = 0$$

- Conservation de la quantité de mouvement, équation de bilan de la quantité de mouvement :

- Conservation de l'énergie : ... **Conservation de la quantité de mouvement (équation de NavierStokes) :**

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{f}$$

Conservation de l'énergie :

$$\rho \left(\frac{\partial e}{\partial t} + \mathbf{u} \cdot \nabla e \right) = -p \nabla \cdot \mathbf{u} + \Phi$$

équations de Navier-Stokes incompressibles, isothermes, en coordonnées cartésiennes :

• **Conservation de la masse (équation de continuité) :**

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

• **Conservation de la quantité de mouvement ($\mathbf{x}, \mathbf{y}, \mathbf{z}$) :**

$$\begin{aligned}\rho \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) &= -\frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) \\ \rho \left(\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \right) &= -\frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) \\ \rho \left(\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right) &= -\frac{\partial p}{\partial z} + \mu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right)\end{aligned}$$

Annexe des opérateurs grad, div, nabla, rot, laplacien, laplacien vectoriel, etc...

Les solutions numériques ne sont pas démontrés

$$(\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u} - \underbrace{\xi \nu \mathbf{D}(\mathbf{u} - \mathbf{u}_{ib})}_{\mathbf{f}_P} \quad (2.1)$$

with $[p] = M \cdot L^{-1} \cdot T^{-2}$ (formalisme divisé par la masse volumique ρ . Donc $[\mathbf{u}] = L \cdot T^{-1}$ - en incompressible).

$$\xi = \begin{cases} 0 & \text{if } \mathbf{x} \in \Omega_f \\ 1 & \text{if } \mathbf{x} \in (\Sigma_b \cup \Omega_b) \end{cases} \quad (2.2)$$

see Figure 2.5

When $D \rightarrow \infty$, problèmes

mais des valeurs trop élevées de D peut mener à des instabilités <- A mentionner qqpart

Fluid mechanics is governed by the NS equations :

- Equations (Incompressible, turbulence,...) - Ajout IBM2.5 -> Methodo (forçage continue ici (vs interpolation ?)) -> avantages - Ajout MRF

Darcy's law in the body interface region Σ_b and the solid region Ω_b . To do so, a spatial dependent tensor $\mathbf{D}(\mathbf{x})$ must be defined. The resulting penalty term is modelled as follows:

$$\mathbf{f}_P = \begin{cases} \mathbf{0} & \text{if } \mathbf{x} \in \Omega_f \\ -\nu \mathbf{D}(\mathbf{u} - \mathbf{u}_{ib}) & \text{if } \mathbf{x} \in (\Sigma_b \cup \Omega_b) \end{cases} \quad (2.3)$$

$\mathbf{u}_{ib}(\mathbf{x}, t)$ is a target velocity representing the immersed body's physical behaviour. ... $\mathbf{u}_{ib} = \mathbf{0}$. The components D_{ij} of the tensor $\mathbf{D}$ are usually controlled to obtain the best compromise between the accuracy and stability of the numerical solver

$$Re_\Omega = \frac{R_2^2 \Omega}{\nu} = 5.53 \times 10^5$$

$$Ma = \frac{R_2 \Omega}{c} = 0.095 \text{ (incompressible)}$$

Reprendre ma présentation (dimensiosn de p)

$$\nabla \cdot (\mathbf{u}\mathbf{u}) = -\nabla p + \nu \nabla^2 \mathbf{u} - \underbrace{\xi}_{\eta} \nu \mathbf{D}(\mathbf{u} - \mathbf{u}_{ib})$$

$$\xi = \begin{cases} 0 & \text{if } \mathbf{x} \in \Omega_f \\ 1 & \text{if } \mathbf{x} \in (\Sigma_b \cup \Omega_b) \end{cases}$$

2.2.2 The open source code OpenFoam

To implemente the IBM forage, we use the open source code OpenFoam, which is a C++ library for computational fluid dynamics (CFD). OpenFoam provides a wide range of tools for solving the Navier-Stokes equations, including solvers, meshing tools, and post-processing tools. It is widely used in the CFD community and has a large user base.

More precisely, for example, to implemente the IBM forage on the speed U , we add the forcing term f_P as follows:

```

1 // Momentum predictor step
2
3 MRF.correctBoundaryVelocity(U) // Corrects U at boundaries to include MRF effects
4                               // (e.g., rotating walls)
5
6 tmp<fvVectorMatrix> tUEqn
7 (
8     fvm::div(phi, U)           // Convective term
9     + MRF.DDt(U)               // Time derivative in the rotating reference frame
10    + turbulence->ddivDevSigma(U) // Turbulent diffusion term
11    + cmptMultiply(D, U - UIB)  // Forcing term: D * (U - UIB)
12    ==
13    fvModels.source(U)          // Source terms from external models
14                                // (e.g., body forces), here equal to 0
15 );
16 fvVectorMatrix& UEqn = tUEqn.ref();
17
18 UEqn.relax() // Applies under-relaxation for numerical stability
19
20 fvConstraints.constrain(UEqn); // Applies physical constraints
21                               // (e.g., boundary conditions)

```

With - *cmptMultiply* the forward product - $D \rightarrow \text{esilon} \times \text{viscosity} \times D - U$ the velocity of the field - UIB the velocity of the walls (here ωR)

```

// Momentum predictor

MRF.correctBoundaryVelocity(U);

tmp<fvVectorMatrix> tUEqn
(
    fvm::div(phi, U)
    + MRF.DDt(U)
    + turbulence->divDevSigma(U)
    + cmptMultiply(D, U - UIB)
    ==
    fvModels.source(U)
);
fvVectorMatrix& UEqn = tUEqn.ref();

UEqn.relax();

fvConstraints.constrain(UEqn);

```

Figure 2.10: implicit forcing term in OpenFoam

To compute the time step Δ_t of the pimpleFoam simulation of the pump, we can use the

Courant-Friedrichs-Lewy (CFL) condition, which states that:

$$\Delta_t = \frac{CFL \cdot \Delta_x}{\|U_\infty\|}, \quad (2.4)$$

where:

- CFL is the Courant number, typically set to 0.5 for stability. A way to visualize this is to think of it as the fraction of the distance a wave travels in one time step relative to the distance between two cells in the azimuthal direction at the tip of the impeller. - Δ_x is the distance between two cells in the azimuthal direction at the tip of the impeller in our model, which is given as 1.5 mm or 0.0015 m. - $\|U_\infty\|$ is the magnitude of the velocity at infinity, which can be approximated by the tangential velocity at the tip of the impeller.

$$\|U_\infty\| = \omega \cdot r_{tip} \quad (2.5)$$

where: - ω is the angular velocity in radians per second, - r is the radius at the tip of the impeller. Assuming the rotating speed is given in revolutions per minute (rpm), we can convert it to radians per second:

$$\omega = \frac{1200 \text{ rpm} \cdot 2\pi \text{ rad}}{60 \text{ s}} = 125.66 \text{ rps}. \quad (2.6)$$

Assuming the radius at the tip of the impeller is $r = 0.2566 \text{ m}$ (which is 0.2566m), we can compute the tangential velocity:

$$\|U_\infty\| = 125.66 \text{ rps} \cdot 0.2566 \text{ m} = 32.24 \text{ m/s}. \quad (2.7)$$

Assuming the distance between two cells in the azimuthal direction at the tip of the impeller is $\Delta_x = 0.0015 \text{ m}$ (which is 1.5 mm), we can now compute the time step Δ_t :

$$\Delta_t = \frac{0.5 \cdot 0.0015 \text{ m}}{32.24 \text{ m/s}} = 2.3 \times 10^{-5} \text{ s} \quad (2.8)$$

2.2.3 EnKF

Préparer l'introduction des variables du controlDict et le CONESDict

2.2.4 The library CONES

CONES is a library coded in Python/C++ developed for ... since 2022 at LMFL. -> appuyer sur "seule online pour CFD à la connaissance du labo" It's parallelized and can be used on supercomputers. It is mainly designed to work with OpenFoam and DA codes as EnKF.

Simulations are performed using the C++ framework OpenFOAM, used to discretize the equations previously presented. Standing for *Open Source Field Operation and Manipulation*, OpenFOAM is an open-source which provides many tools, referred to as executables

- **Pre-processing**, including meshing and decomposing tools.
- **Solving**, including in-built and custom solvers. Each solver is built to answer a specific CFD problem.
- **Post-processing**, including ParaView, an open-source visualization application.

OpenFOAM is not provided with a graphical interface, which is why a simulation is built by writing "dictionaries", in which are stored the necessary information for the simulation (mesh,

turbulence model, maximum number of characteristic times, ...). Thus, a specific directory structure must be observed, as outlined in ...

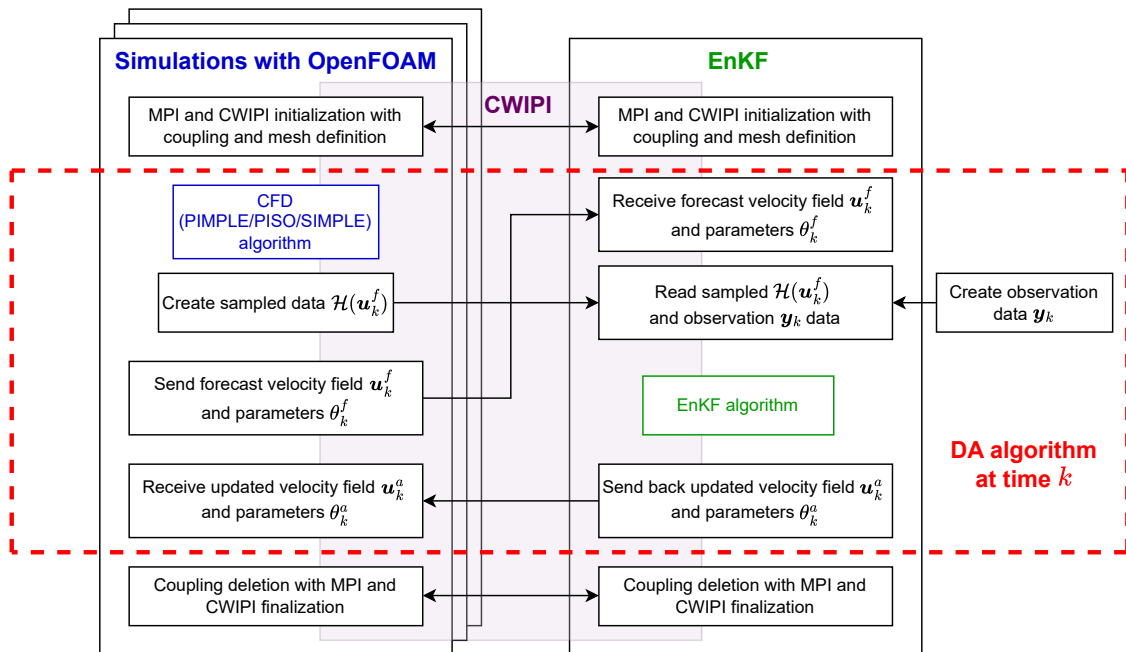


Figure 2.11: CONES

```

root
├── system
│   ├── controlDict
│   ├── blockMeshDict
│   ├── decomposeParDict
│   ├── fvSchemes
│   └── fvSolution
├── templates
└── tests
    
```

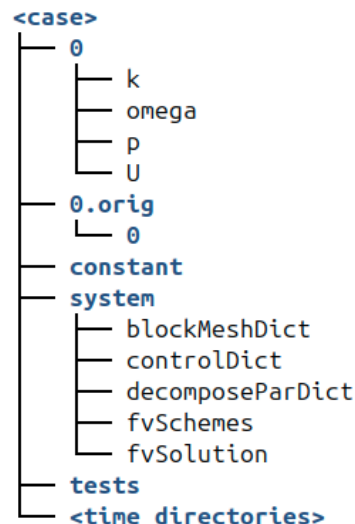


Figure 2.12: Arbo of OpenFoam

The **system** directory contains the main parameters concerning the progress of the simulation:

- The **controlDict** dictionary defines parameters such as start and end time, time steps, data output formatting, or data average computation.
- The **blockMeshDict** dictionary defines the geometry, mesh and boundary conditions implementation.
- The **decomposeParDict** dictionary defines the geometry and mesh decomposition, with the objective to run the simulation in parallel, using multiple cores.
- The **fvSchemes** dictionary defines the discretisation schemes used in the solution.
- The **fvSolution** dictionary defines the equation solvers, tolerances and other algorithm controls for instance.

The **constant** directory contains the different physical properties of the simulation. Moreover, the subfolder **polyMesh** contains the mesh data, obtained after the **blockMesh** command line is run:

- The **momentumTransport** dictionary defines the simulation type (RANS or LES), as well as the turbulence model.
- The **transportProperties** dictionary defines the transport model (Newtonian for instance) and some transport coefficients, such as the kinematic viscosity ν .
- The **turbulenceProperties** dictionary defines the different parameters of the chosen turbulence model.

The **0** directory contains the initial solution's fields. The **postProcessing** directory stores data from the running simulation, such as forces and probes' data. The **processor0...N** directories directly result from the geometry decomposition, which is proceeded through the **decomposeParDict** dictionary. The **dummy.foam** is an empty file which allows to visualize the simulation with ParaView.



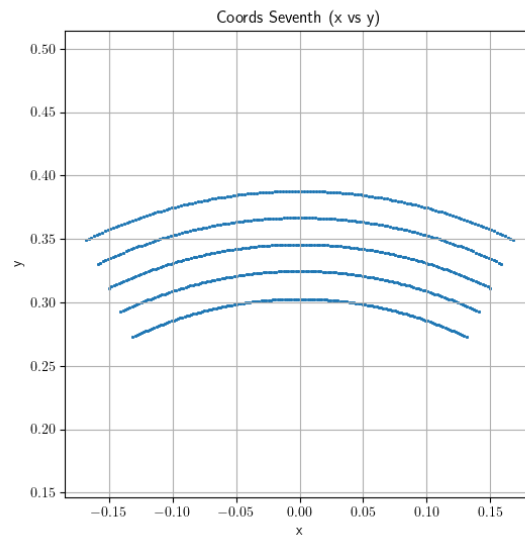


Figure 2.14: Interpolation points over 360/7 degrees

Notre but : faire fit ces 2 courbes :

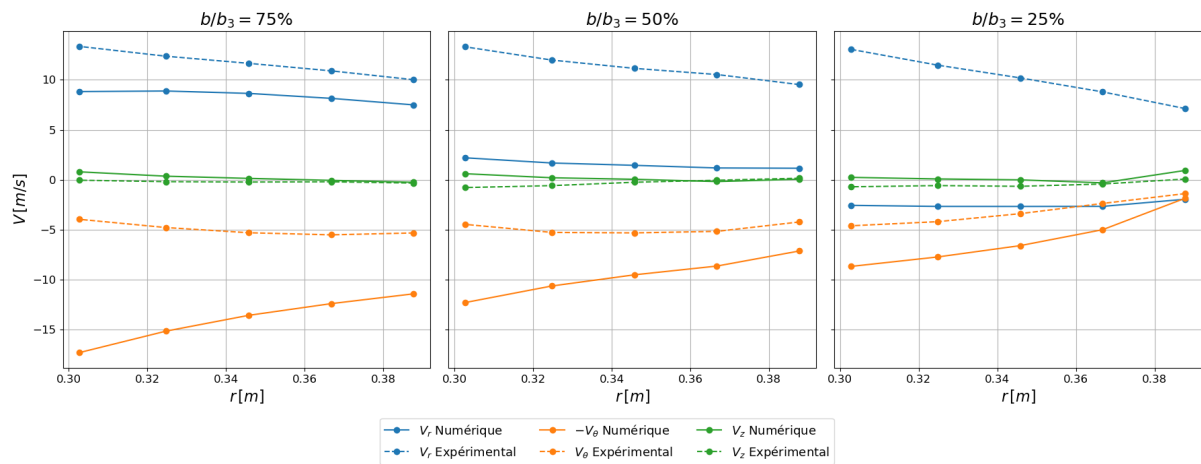


Figure 2.15: implicit forcing term in OpenFoam

Mais surtout obtenir le bon rendement ! (sujet du stage)

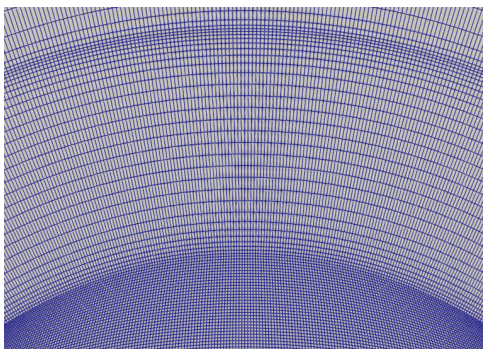


Figure 2.16: diffuseur Mesh

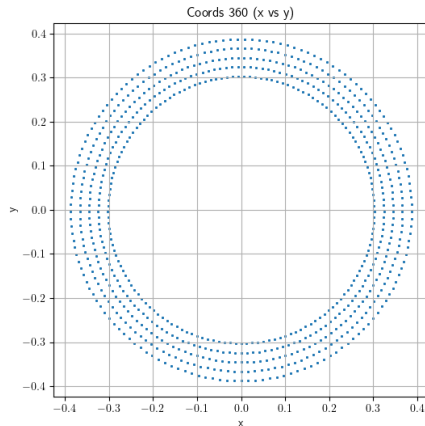


Figure 2.17: Interpolation points over 360 degrees

Limitations IBM à améliorer avec DA

2.4 Application of the DA

Explication de coords_seventh.txt, coords_360.txt et coords_7.txt

Tentative d'explication en français :

J'ai d'un coté une simulation numérique de la pompe centrifuge, qui est un modèle numérique de la pompe. De l'autre coté, j'ai des données expérimentales de PIV, qui sont des données expérimentales de la pompe centrifuge. Je veux utiliser les données expérimentales pour améliorer le modèle numérique de la pompe centrifuge. Pour cela, j'utilise la méthode de l'assimilation de données, qui est une méthode qui permet d'améliorer un modèle numérique en utilisant des données expérimentales. La méthode que j'utilise est l'Ensemble Kalman Filter (EnKF), qui est une méthode d'assimilation de données qui utilise un ensemble de modèles numériques pour améliorer le modèle numérique. Cela se fait en deux étapes :

1. Je fais une simulation numérique de la pompe centrifuge, en utilisant le modèle numérique de la pompe centrifuge. Cette simulation numérique me donne un ensemble de modèles numériques, qui sont les membres de l'ensemble.
2. Je compare les données expérimentales de PIV avec les données numériques de la simulation numérique. Pour cela, je fais une interpolation des données expérimentales de PIV sur les points de l'ensemble de modèles numériques. Cette interpolation me donne un ensemble de données expérimentales, qui sont les observations.

La comparaison entre les données expérimentales et les données numériques me permet de calculer l'écart entre les deux, qui est appelé l'innovation (mettre référence partie mathématique). L'innovation est utilisée pour mettre à jour les membres de l'ensemble de modèles numériques, en utilisant la méthode de l'EnKF. Cette mise à jour me permet d'améliorer le modèle numérique de la pompe centrifuge.

1. Ma simulation numérique de la pompe est stationnaire en utilisant la méthode MRF. Cela signifie que le champ de vitesse est constant dans le temps. Cependant, il n'est pas constant dans l'espace, car il dépend de la position dans le diffuseur. Un motif de vitesse répété sept fois est donc censé être visible (cela n'est pas vraiment le cas en instantané).

2. Les données PIV sont acquises à une vitesse de rotation de la pompe et une fréquence telle que 49 images sont acquises par tour de roue, soit 7 images entre deux passages d'aubes

successives. Dans le référentiel de la pompe, les images sont prises à une différence de $360/49$ deg.

Nous voulons comparer des données consistantes. Nous pourrions synchroniser une simulation instationnaire avec les données PIV. Mais nous avons choisis de faire une simulation instationnaire dans un premier temps.

Nous pourrions reconstruire le champ spatial PIV (nous avons assez d'angle de vue et de données pour cale -> ref article Antoine). Cependant, la vitesse de rotation n'est pas très précise, mais surtout, la méthode MRF inverse la forme du sillage.

Nous avons donc choisis de moyenner temporellement les données PIV (ce qui revient à moyenner spatialement). Du côté de la simulation, nous avons donc besoin de moyenner azimuthalement les données numériques pour les rendre compatibles avec les données PIV.

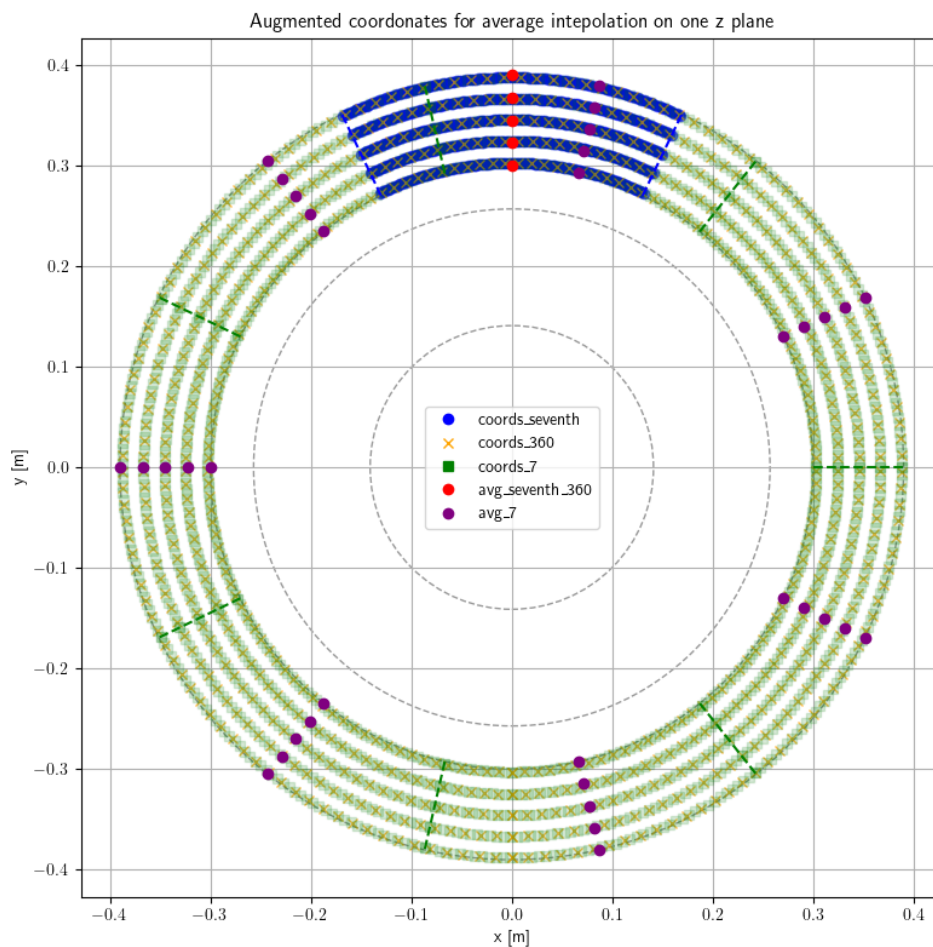


Figure 2.18

Imager avec des illustrations et vue en coupe ParaView

Pour cela j'ai commencé en moyennant $1/7$ mais les paramètres étant locaux (non globaux), j'ai ensuite moyenné sur 360 degrés, ce qui permet de mieux prendre en compte les variations locales des paramètres. Cependant, nous pouvons converger vers une moyenne physique qui atteint un minimum local mais qui ne respecte pas d'avoir 7 fois le même pattern. Cela permet à F_x d'être très différent de F_y mais aussi à la simulation de diverger avec des paramètres et vitesses erronées. Pour palier ce problème, je compare la moyenne des patterns sur mes 7 segments de $360/7$ degrés.

Nous avons fait l'EnKF sans l'état... Sans inflation pour commencer. Si on converge vers une

solution qui ne nous satisfait pas, on peut essayer d'ajouter... Cela permet de s'échapper d'un minimum local en ajoutant du bruit artificiellement.

Nous avons fait une simu RANS mais nous aurions pu faire une simu URANS (ou LES ou DDES) avec un modèle de turbulence...

L'expérience montre un ordre de grandeur de centaines d'itération entre les phases d'analyse, mais cela dépend beaucoup du cas. Il en est de même pour le nombre de phases d'analyse. Supposons 1000 itération en moyenne entre chaque phase d'analyse, et 300 phases d'analyse, cela nous fait 300 000 itérations à simuler, soit 93h, sur env. 2800 coeurs, soit 264kh.processeur de calcul.

Comment calculer la variance que je veux mettre ? Expérience : 5% De mon coté : je vise qu'en moyenne un membre d'ensemble ait un écart de $1e7$ par rapport au temps précédent (pour des valeurs autour de $5e7$), donc, pour 30 membres d'ensemble, environ $95\% = 2 \text{ sigma}$ dans $5e7 \pm 1e7$, càd $\text{sigma} = 0.5e7 = 10\%$. Ensuite, le nombre d'itération entre les phases d'analyse est déterminé en regardant combien d'itération sont nécessaires pour faire varier la solution dans le diffuseur (influe, rapidement, converge doucement mais ce qui nous intéresse c'est que ça prenne la tendance) lorsqu'un paramètre de forçage est modifié (supposé très inférieure pour la turbulence car appliquée de partout)

The observation window was defined by observing the number of iterations required for the influence of the IBM to reach the diffuser and take the form of the converged result. We are not interested in the turbulence parameters to determine the observation window because they are present throughout the domain, their influence is almost direct (not necessarily their convergence). We chose 800 iterations

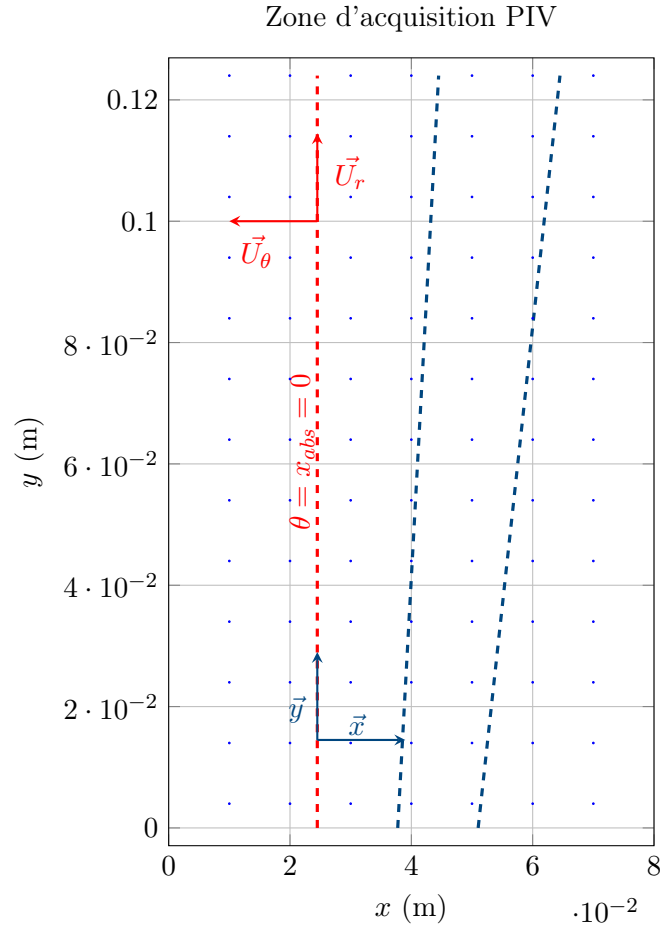
Pb de la DA : plus rapide que DNS mais toujours plus lent que RANS et le temps d'attente et de calcul sur supercalculateur (sur 4000 processeurs) est de l'ordre de plusieurs jours. On peut donc se demander si la DA est vraiment utile dans ce cas. En effet, la DA est plus rapide que la DNS mais pas que la RANS. Cependant, il faut prendre en compte le fait que la DA permet de réduire le temps de calcul de la simulation numérique en permettant d'optimiser les paramètres de l'IBM. De plus, la DA peut être utilisée pour d'autres cas d'étude où la DNS est trop coûteuse.

Pb perso : jusqu'à plusieurs jours d'attente pour lancer une simu.

2.4.1 Preprocessing of PIV data

PIV acquisition treatment

The PIV data is acquired in a rectangular zone of $0.08 \text{ m} \times 0.124 \text{ m}$. The cartesian coordinates are given in a local system, where the absolute origin is at the center of the pump.



Dans la simulation numérique, l'axe z est vers le centre de la Terre, contrairement à l'expérience où il est vers le haut. La roue tourne dans le sens trigonométrique dans le repère de la simulation numérique et dans le sens antitrigonométrique dans l'expérimentation. Actuellement, je respecte le formalisme de la simu num pour le repère.

Parler des coefficients de relaxation.

1. First, we need to know how to convert the coordinates from the local system to the absolute system. We find that between the line 55 and 56, the radius is constant and pass from lower than π to upper. In fact, for the plan at 75%, it is the same approach but between the line 65 and 66 because of some technical experimental constraints. We also did plot x over theta for three different values of theta and find out an intersection point for $x = 0.0245$. So now, we know that for $x = 0.0245$, $\theta = \frac{\pi}{2}$

$$\begin{cases} x_{rel} = x_{abs} + 0.0245 \\ y_{rel} = x_{abs} + R_2 \end{cases} \quad (2.9)$$

2.Average For now we are in stationary case (time invariant). So we are going to deal with data average over time. There are around 49 maps (=file) by turn. So to average the data over the time, we can average over the ϕ (ϕ appartient à 0-211rd) (=maps). That into the six directory.

3.Sampling We will also average data between R1 and R2 to go to 3 files (of 10x81x125 elements). Now, we keep only the data on the line $x = 0.0245$ (in fact we average data between x

= 0.024 and $x = 0.025$). Data are then truncate at line 89, where the radius is upper than 0.3, because of some laser reflexion on the pump during the PIV experiment.

4. Shaping for CONES Cones keep data in a particular way. We need to convert the data in a format that cones can read. From 3 files (of 10x81x125 elements), to the files `fieldCylindrical.txt` and `obs_coordinates.txt`. Describe shapes...

5. Creation of a file for the interpolation Finally, we need to create an `coords_seventh.txt` file for the interpolation function in CONES. We don't just use the coordinates of the points in the PIV data, because we want to average over the angle θ on the pump. The numerical simulation is $\frac{2\pi}{7}$ OU $2\pi/7$ periodic, so we interpolate in 150 points over theta between 0 and $2\pi/7$. To do that, we just take the `obs_coordinates.txt` file and duplicate 150 times the points over the angle θ (between $-\pi/7$ and $\pi/7$). In our case, we chose 150 that will approximately correspond to the shape of a cell at this radius. To be consistent with axis further, we take points between $-\pi/7 + \pi/2$ and $\pi/7 + \pi/2$

Here is the plot of the experimental data and the numerical data (for $D = 3e7$ (introduire la variable)), at the same points (both are averages over time and over the angle θ).

potentiellement en annexe

Concretely, to average the data over the angle θ , we need to modify the interpolation function in CONES. Instead of read `coords`, the interpolation function read the data in the `coords_seventh.txt` file.

Then, it computes the interpolation and average the data over the angle θ (between $-\pi/7$ and $\pi/7$), by doing a somme over θ and dividing by the number of points (150). To go back from 150x5x3x3 (to detail) to 5x3x3 data. All of those computation are mad in a cylindrical referencial. Ref au schéma tikz

A sixth coefficients DP has been add. It's represent the difference of pressure between the inlet of the pipe and the outlet of the diffuser. The numerical coefficient are interpolating, respectively at (0,0,-0.35) and (0,0.390,0). For the high fidelity value, we can find experimental data form the same pump [M. Fan 2022], that's match with M. Fan simulation (eq to approximately $\eta = 0.4$). So can also compute :

U_2 is the impeller tip speed computed from the impeller diameter and the rotation speed of the pump.

$$\frac{\Delta P_P}{\rho U_2^2} = \eta \Leftrightarrow \Delta P_P = \rho U_2^2 \eta \quad (2.10)$$

$$U_2^2 = (\omega * R_2)^2 = (2 * \pi * 1200 / 60 * 0.2566)^2 = 32,2 \text{ m/s}$$

As we are in a steady simulation, the result of the numerical pressure is divided by the density of the fluid.

$$\Delta P_P = 32.2^2 \times 0.4 = 415 Pa \quad (2.11)$$

So the state vector is now 46. Incertitude on P will be less to take more take it account. Inflation...

At pump flow rate = pump design flow rate So $Q^*/Q^* = 1$

2.4.2 Modifications on OpenFoam and CONES

Tout ce qu'il y a dans le CONES dict Parler des modifs dans le code ici.

- Reduction of the pipe (with the research of the velocity inlet condition)

Toujours mentionner les paramètres utilisés (dans le controlDict et le CONESDict)

2.4.3 Application of the DA

2.5 Results

Comparer les 3 résultats : avant DA, après simple et "complex" DA

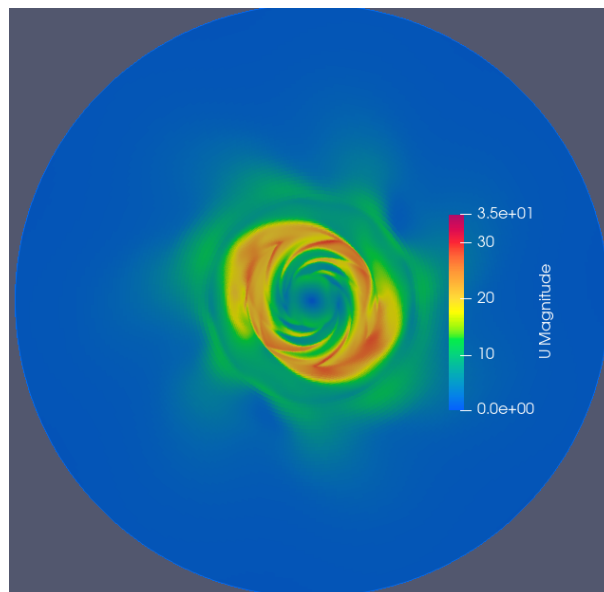


Figure 2.19: Velocity field for $Dx = 4e7$ and $Dy = 2e7$

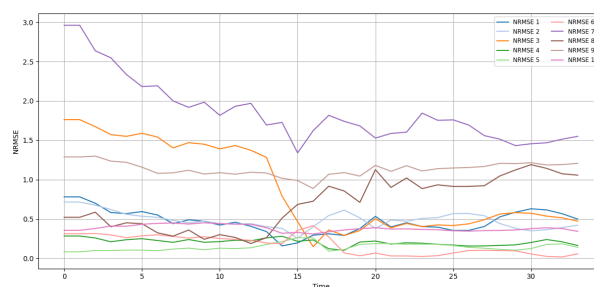


Figure 2.20: Evolution of the NRMSEs over EnKF analysis phases

...

Explication des écarts par différentes choses

Tt ce que j'ai implémenté en plus (déjà détaillé partie précédente normalement)

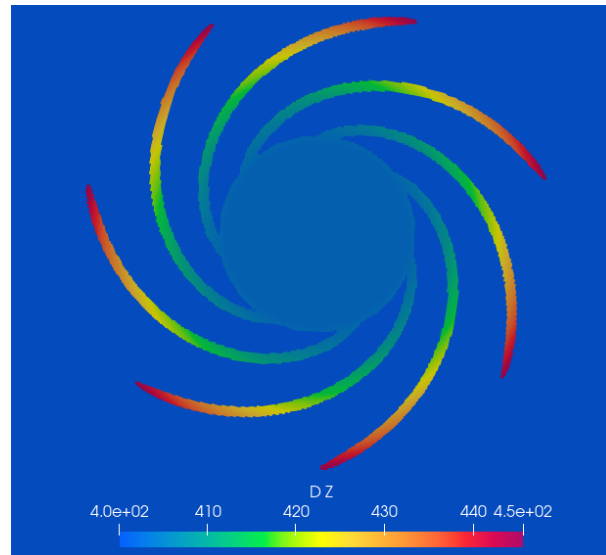


Figure 2.21: Radial expansion of Dz over the impeller radius

Essayer d'améliorer le rendu de l'image ci-dessous en mettant la pompe en transparence

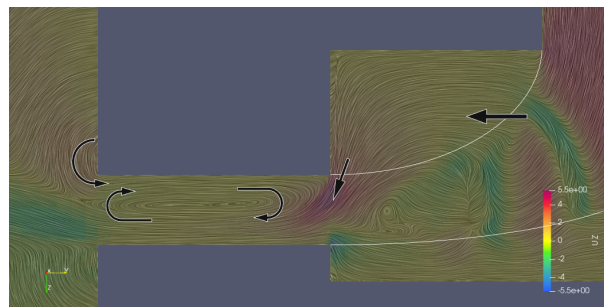


Figure 2.22: Formation of the recirculation bubble in the diffuser

Explication des écarts par la bulle de recirculation.

2.5.1 Conclusion ans perspective

Problème de méthode et pas de miracle ave cette méthode IBM. Interpolation, instationnaire (avantage : on ne contraint plus avec MRF et donc plus facile de réconcilier solution numérique physique et mathématique)

Ouverture : méthode IBM interpolation, article non référencé.

[9] -> ML turbulent channel flow Miguel

"Ouverture / idée" : DA temps réel dans avion impossible pour encore longtemps. Solution : Contrôle actif de la puissance en fonction des mesures de pression dans le compresseur : $f(\text{obs}) \rightarrow$ controle de la puissance Pour ça, se ref au sujet de thèse de Tarane obs pression $\rightarrow$ modèle de reconstruction du champ de vitesse $\rightarrow$ modèle de reconnaissance de patternes de décrochage $\rightarrow$ contrôle de la puissance Peut-être faire deux filtres de Kalman imbriqués ? hyperparamètres type fenetre observation, variance initiale, ...

Voir "TD -> Done et "TDSave -> Done" pour voir tt ce que j'ai fais

The random variations are random but with a reproducibility seed (sed) that allows to have the same result each time the simulation is launched.

Conclusion

Technical and human review

Vérifier les consignes

Outlook

Personal reflection

Annexes

Batch code used to run the simulation on the Rome partition of TGCC cluster.

```
#!/bin/bash                                # To be run with bash
#MSUB -r LCP35                             # Job name
#MSUB -o LCP35_%I.out                     # Output name. %I is the job ID
#MSUB -e LCP35_%I.err                     # Error name. %I is the job ID
#MSUB -q rome                             # Partition
#MSUB -A gen1741                           # Project ID
#MSUB -n 4446                             # Cores nb (35 members x 127 OF) + 1 python
##MSUB -c 2                               # To allocate 2 cores per task (default is 1)
##MSUB -N 1                               # Nb of calcul node (128 cores per node on rome)
#MSUB -T 43200                             # Elapsed time limit in seconds
##MSUB -Q long                            # (upto 3 days)
##MSUB -Q test                            # (short, high priority jobs)
#MSUB -m scratch                           # Execute in scratch directory
#MSUB -@ name@mail.fr:begin,end           # Email address

set -x
cd ${BRIDGE_MSUB_PWD}
module purge
module load gnu/8 mpi/openmpi/4 flavor/python3/cuda python3/3.10.6 openfoam/9
source $FOAM_SOURCE_FILE

./visuNParams.sh                           # Script .sh
ccc_mprun -f app_mySF35.conf               # Run the application with the configuration file
```

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List of acronyms

AI	Artificial Intelligence
CERFACS	Centre Européen de Recherche et de Formation Avancée en Calcul Scientifique
CFD	Computation Fluid Dynamics
CFL	Courant-Friedrichs-Lewy
CHSCTN	Comité d'Hygiène, de Sécurité et des Conditions de Travail National
CNRS	Centre National de la Recherche Scientifique
CONES	Coupling OpenFoam with Numerical EnvironmentS
CPER	Contrat de Plan État-Région
CSE	Comité Social et Économique
CSR	Corporate Social Responsibility
CTI	Commission des Titres d'Ingénieur
DA	Data Assimilation
DNS	Direct Numerical Simulation
EnKF	Ensemble Kalman Filter
ENSAM	École Nationale Supérieure d'Arts et Métiers
HCERES	Haut Conseil de l'Évaluation de la Recherche et de l'Enseignement Supérieur
IBM	Immersed Boundary Methods
IPSA	Institut Polytechnique des Sciences Avancées
JFM	Journal of Fluid Mechanics
LES	Large Eddy Simulation
LMFL	Laboratoire de Mécanique des Fluides de Lille
ONERA	Office National d'Études et de Recherche Aéronautiques
PIV	Particle Image Velocimetry
RANS	Reynolds-Averaged Navier-Stokes equations
RSE	Responsabilité Sociétale des Entreprises
RSU	Rapport Social Unique
SHF	Société Hydrotechnique de France

Résumé Vérifier les consignes

Abstract